

1 Definitions of Effective World Model and Empirical World Model

Large Language Models (LLMs) have shown strong performance in various tasks, which indicates that they possess some form of internal *world model* to understand and reason about the world. We use the Bayesian framework as an external **analytical tool** to interpret and diagnose this behavior.

1.1 Effective World Model

We do not assume the LLM mechanistically performs Bayesian calculations. Instead, we model its complete input-output behavior using the Bayesian notations. This "Effective World Model" is the theoretical, unobservable system that would perfectly replicate the LLM's complete behavior. We can conceptualize its components as:

- **Effective Prior** ($p_{\text{true}}(\theta)$): The LLM's latent, "true" belief about θ before it is given specific evidence. We might probe this by asking generic questions (e.g., "What is the capital of France?").
- **Effective Likelihood** ($f_{\text{true}}(y|\theta)$): The LLM's implicit model of how evidence y relates to hypothesis θ . It captures which observations the LLM implicitly expects to see if a particular hypothesis were true.
- **Effective Posterior** ($q_{\text{true}}(\theta|y)$): The LLM's latent, "true" final belief state after actually processing the evidence y . (This is distinct from the analytical $p_{\text{true}}(\theta|y)$, which would be the Bayesian update of p_{true} and f_{true}).

This model is a theoretical construct that we cannot observe directly.

Note that we make **no assumption** that $q_{\text{true}}(\theta|y)$ follows Bayes' rule. The relationship between $q_{\text{true}}(\theta|\emptyset)$ and $q_{\text{true}}(\theta|y)$ characterizes how the LLM actually updates its beliefs, which may or may not be Bayesian.

1.2 The "Approximate" (Empirical) World Model

This model consists of the quantities we *empirically measure* from the LLM using various Uncertainty Quantification (UQ) estimation methods (verbalized prompting, logits, and sampling etc). These are our "approximate" measurements of the Effective World Model components.

Approximate Prior ($p(\theta)$): Our empirical estimate of the LLM's effective prior belief, obtained via UQ methods.

Approximate Likelihood ($f(y|\theta)$): Our empirical estimate of the LLM's effective likelihood function.

Approximate Posterior ($q(\theta|y)$): Our empirical measurement of the LLM’s final belief state for θ after it has processed the evidence y .

Model Evidence ($p(y)$): The marginal likelihood *derived from our approximate components*.

$$p(y) = \int p(\theta)f(y|\theta) d\theta \quad (1)$$

This value represents the total probability of observing the data y , given our approximate model.

Analytical Posterior ($p(\theta|y)$): The theoretical posterior distribution derived from our approximate prior $p(\theta)$ and likelihood $f(y|\theta)$ via Bayes’ Theorem. **This is our ”target” distribution.**

$$p(\theta|y) = \frac{p(\theta)f(y|\theta)}{p(y)} = \frac{q_0(\theta)}{p(y)} \quad (2)$$

A note on interpretation: a framework, not a mechanism. It is crucial to state that this Bayesian framework is an **analytical tool** for interpretation, not a **mechanistic description** of how a Large Language Model (LLM) operates. Mechanistically, a transformer is a feed-forward network that maps a sequence of input tokens to a probability distribution over the next token. It does not contain literal, separate components for ”prior” and ”likelihood” that it combines using Bayes’ rule. Instead, we are **modeling** the LLM’s observable behavior **as if** it were a Bayesian agent.

1.3 Internal and External Consistency

Definition 1 (True Internal Consistency). Let the effective components be the prior $p_{\text{true}}(\theta)$, likelihood $f_{\text{true}}(y | \theta)$, and posterior $q_{\text{true}}(\theta | y)$, where

$$p_{\text{true}}(\theta | y) \propto p_{\text{true}}(\theta) f_{\text{true}}(y | \theta).$$

Define the *latent internal inconsistency* at observation y as:

$$C_{\text{true}}(y) := D_{\text{KL}}(q_{\text{true}}(\theta | y) \| p_{\text{true}}(\theta | y)).$$

The model is internally consistent if and only if $C_{\text{true}}(y) = 0$ for all y . *These quantities are unobservable.*

Definition 2 (Empirical Internal Consistency). From elicited approximations p, f, q , define:

$$C(y) := D_{\text{KL}}(q(\theta | y) \| p(\theta | y)) \geq 0.$$

This measures how empirically internally consistent the model behaves. Internal consistency concerns empirical posterior relative to **analytical** posterior calculated from prior and likelihood.

Definition 3 (External Consistency). External consistency concerns empirical accuracy relative to **ground truth or the real task**. For tasks with ground-truth posterior $p^*(\theta | y)$ and model predictions $q(\theta | y)$, define the *external inconsistency* as:

$$E_{\text{ext}} := \mathbb{E}_{y \sim \mathcal{D}} [D_{\text{KL}}(p^*(\theta | y) \| q(\theta | y))].$$

Large values of E_{ext} indicate poor alignment with ground truth. External consistency measures empirical accuracy independent of the model's internal coherence.

2 Zellner's Information-Theoretic Framework

Output Information (I_{out}):

$$I_{\text{out}} = \int q(\theta|y) \log q(\theta|y) d\theta + \log p(y) \quad (3)$$

I_{out} is the **negative entropy** of the posterior distribution $q(\theta|y)$ **plus the log-evidence** $\log p(y)$. (Since Entropy $H(p) = -\int p \log p$, $I_{\text{out}} = -H(p(\theta|y)) + \log p(y)$).

Input Information (I_{in}): The input information I_{in} is a functional of the approximate posterior $q(\theta|y)$.

$$I_{\text{in}}(q) = \int q(\theta|y) \log q_0(\theta) d\theta = \int q(\theta|y) \log[p(\theta)f(y|\theta)] d\theta \quad (4)$$

$I_{\text{in}}(q)$ is the **negative cross-entropy** between the approximate posterior $q(\theta|y)$ and the (unnormalized) input information $q_0(\theta)$. (Since Cross-Entropy $H(q, q_0) = -\int q \log q_0$, $I_{\text{in}}(q) = -H(q, q_0)$).

Information Loss Functional ($G(q)$): We minimize the information loss, which is the difference $I_{\text{out}} - I_{\text{in}}$.

$$G(p) = I_{\text{out}} - I_{\text{in}} \quad (5)$$

$$= \int q(\theta|y) \log q(\theta|y) d\theta + \log p(y) - \int q(\theta|y) \log q_0(\theta) d\theta \quad (6)$$

$$= \int q(\theta|y) \log \frac{q(\theta|y)}{q_0(\theta)} + \log p(y) \quad (7)$$

$$= \int q(\theta|y) \log \frac{q(\theta|y)}{p(\theta)f(y|\theta)} + \log p(y) \quad (8)$$

$$= \int \int q(\theta|y) \log \frac{q(\theta|y)}{p(\theta|y)} \quad (9)$$

$$= D_{KL}(q(\theta|y) \| p(\theta|y)) \quad (10)$$

where the analytical posterior is $p(\theta|y) = \frac{p(\theta)f(y|\theta)}{p(y)}$.

Optimal Posterior: The KL divergence $D_{KL}(q(\theta|y) \| p(\theta|y))$ is minimized at 0 if and only if $q(\theta|y) = p(\theta|y)$. Therefore, the optimal posterior that conserves information is:

$$q_{\text{optimal}}(\theta|y) = \frac{p(\theta)f(y|\theta)}{p(y)} = p(\theta|y) \quad (11)$$

This is precisely Bayes' Rule.

3 Information Gain and Log-Likelihood Ratio

The ****Information Gain**** measures the change in information from the prior $p(\theta)$ to the analytical posterior $p(\theta|y)$, defined as the KL divergence between them.

Core Idea: Under optimal Bayesian updating, the KL divergence from prior to posterior equals the expected log-likelihood under the posterior minus the log marginal likelihood. For binary hypothesis testing with equal priors, this simplifies to approximately the log-likelihood ratio when one hypothesis dominates.

Information Conservation: Zellner's principle states that the information gain (KL from prior to posterior) should equal the information content of the data (log-likelihood ratio), establishing:

$D_{KL}(\text{posterior} \ \text{prior}) = \text{Information from Data} = \text{LLR (for optimal processing)}$

(12)

$$\text{Information Gain} = D_{KL}(p(\theta|y) \| p(\theta)) \quad (13)$$

$$= \int p(\theta|y) \log \left[\frac{p(\theta|y)}{p(\theta)} \right] d\theta \quad (14)$$

From our definition of the Analytical Posterior in Section 1, we can rearrange Bayes' Rule:

$$\frac{p(\theta|y)}{p(\theta)} = \frac{f(y|\theta)}{p(y)} \quad (15)$$

We substitute this ratio into the Information Gain equation:

$$\text{Information Gain} = \int p(\theta|y) \log \left[\frac{f(y|\theta)}{p(y)} \right] d\theta \quad (16)$$

We can split the logarithm:

$$= \int p(\theta|y) (\log f(y|\theta) - \log p(y)) d\theta \quad (17)$$

$$= \int p(\theta|y) \log f(y|\theta) d\theta - \int p(\theta|y) \log p(y) d\theta \quad (18)$$

The first term is the posterior expectation of the log-likelihood. The second term simplifies because $\log p(y)$ is a constant and $\int p(\theta|y) d\theta = 1$:

$$= E_{p(\theta|y)}[\log f(y|\theta)] - \log p(y) \quad (19)$$

This provides the fundamental identity:

$$D_{KL}(p(\theta|y) \| p(\theta)) = E_{p(\theta|y)}[\log f(y|\theta)] - \log p(y) \quad (20)$$

This shows that the Information Gain is exactly the difference between the posterior-expected log-likelihood and the log-evidence.

3.1 Approximate Gain vs. Approximate LLR

We now analyze the relationship between two quantities derived from our **approximate posterior** $q(\theta|y)$.

1. **Approximate Information Gain:** This measures the information-theoretic "distance" from the prior to our empirical posterior.

$$\text{Approximate Gain} = D_{KL}(q(\theta|y) \| p(\theta)) \quad (21)$$

2. **Approximate LLR:** This is the "Log-Likelihood Ratio" term from Section 6, but evaluated using the expectation under our approximate posterior $q(\theta|y)$.

$$LLR_{\text{approx}} = E_{q(\theta|y)}[\log f(y|\theta)] - \log p(y) \quad (22)$$

We can find the difference between these two terms. This derivation relies on the **Evidence Lower Bound (ELBO)**, a term in a fundamental mathematical identity that holds for any choice of p , f , and q .

The ELBO has two well-known definitions:

$$\mathcal{L}(q) = \log p(y) - D_{KL}(q(\theta|y) \parallel p(\theta|y)) \quad (23)$$

$$\mathcal{L}(q) = E_{q(\theta|y)}[\log f(y|\theta)] - D_{KL}(q(\theta|y) \parallel p(\theta)) \quad (24)$$

We start by rearranging the second ELBO identity (Equation 24) to solve for our Approximate Gain term:

$$D_{KL}(q(\theta|y) \parallel p(\theta)) = E_{q(\theta|y)}[\log f(y|\theta)] - \mathcal{L}(q) \quad (25)$$

Now, we compute the difference of interest by substituting Equation 25 for the Approximate Gain:

$$\text{Approximate Gain} - LLR_{\text{approx}} \quad (26)$$

$$= (E_{q(\theta|y)}[\log f(y|\theta)] - \mathcal{L}(q)) - (E_{q(\theta|y)}[\log f(y|\theta)] - \log p(y)) \quad (27)$$

The $E_q[\log f(y|\theta)]$ terms cancel out, leaving:

$$= \log p(y) - \mathcal{L}(q) \quad (28)$$

Finally, we use the first ELBO identity (Equation 23), which states $\log p(y) - \mathcal{L}(q) = D_{KL}(q(\theta|y) \parallel p(\theta|y))$. This gives us our final result:

$$D_{KL}(q(\theta|y) \parallel p(\theta)) - LLR_{\text{approx}} = D_{KL}(q(\theta|y) \parallel p(\theta|y)) \quad (29)$$

This identity mathematically proves that the difference between the "Approximate Gain" and the "Approximate LLR" is exactly the KL divergence between the approximate posterior $q(\theta|y)$ and the analytical posterior $p(\theta|y)$. This value, $D_{KL}(q \parallel p)$, is our approximation error and is guaranteed to be non-negative.